ARJAN SINGH GILLAN

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EDUCATION

University of California, Irvine — Irvine, CA
B.S. in Biomedical Engineering | GPA: 3.64 — *Graduation: June 2028*

EXPERIENCE

Applications & Clinical Engineering Intern — Irvine, CA
Eyedaptic — AR/AI Vision Assistance for Macular Degeneration — *Jan 2026 – June 2026*

- Ran testing and validation of an AI-driven vision-assistance platform on AR glasses, benchmarking 4 AR headsets and 4 AI agents on response quality and camera performance.
- Designed comparative evaluations of camera image quality across devices, documenting findings to guide product and engineering improvements.

Assistant Project Manager, University Rover Challenge — Irvine, CA
Legacy Robotics at UC Irvine — *April 2026 – Present*

- Oversee an 80-person engineering design team, spanning 5 subteams across multiple engineering majors, building a Mars rover for the 2027 University Rover Challenge.
- Manage a ~$15K budget and ensure the rover meets the mechanical and electrical constraints of the competition across all subsystems.
- Developed an onboard soil biosignature assay to detect protein concentrations as a life indicator and authored the chemical safety and lab-handling protocol for onboard testing.

Undergraduate Research Assistant, Biomechanics — Irvine, CA
UCI Rehabilitation & Augmentation Lab — *April 2025 – Present*

- Conduct human gait rehabilitation trials applying spinal cord stimulation during walking and criss-cross tasks, analyzing 3D motion-capture data in Vicon Nexus to extract joint kinematics and step metrics across baseline and stimulation sessions.
- Use MATLAB to quantify how movement direction and speed affect ankle proprioception, optimizing rehabilitation protocols for stroke-patient recovery.

Undergraduate Research Assistant, Neuroimaging — Irvine, CA
UCI Center for Neural Circuit Mapping — Xu Lab — *Jan 2025 – Present*

- Segment degu brain MRI scans in ITK-SNAP to build a novel neuroanatomical atlas and developed a program to quantify regional brain shrinkage for Alzheimer's research.

TECHNICAL PROJECTS

Bridgeless Eyewear System — Irvine, CA
Computer Aided Design and FEA Project — *May 2026 – June 2026*

- Co-designed a bridgeless eyewear system in SolidWorks that removes the nasal bridge from the load path, redistributing frame weight to the sides and back of the head through adjustable temple arms — for users recovering from nasal surgery or with saddle-nose and congenital nasal conditions.
- Validated a 58.9 g / 0.58 N 3D-printed prototype with static FEA (stress, strain, displacement) and hinge-fold and arm-extension motion studies, incorporating ball-detent sizing and a spring-loaded slide for adjustable, bridge-free fit.

Morse Code Learning Device — Irvine, CA
Data Acquisition Project — *Nov – Dec 2025*

- Built a working Morse code learning device on Arduino in embedded C++: decoded button presses into dots/dashes with buzzer and LED feedback, mapped patterns to characters via a 48-entry lookup (letters, numbers, punctuation), and added two-player messaging plus a serial-to-Morse mode that plays typed messages back audibly.

Autocorrelogram Firing Program — Irvine, CA
Data Analysis Sandbox Project — *Nov – Dec 2025*

- Automated classification of neuronal firing patterns with a custom MATLAB tool, adding ACG visualization of electrophysiological datasets.

LEADERSHIP

Vice President of Finance — Irvine, CA
Engineering Student Council at UCI — *April 2026 – Present*

- Oversee a ~$40K annual budget for a 22-member board, managing allocation and financial planning across council programs and events.

SKILLS & INTERESTS

Technical: Testing & Validation, Data Acquisition & Analysis, MATLAB, C++, C#, Arduino, ITK-SNAP, Vicon Nexus, Histology, PCB Design, Soldering, 3D Printing, CNC, CAD

Interests: Event Planning, Public Speaking, Cooking, Rock Climbing, Golf, Tennis